

Paper IV

Apparatus to printed pp. 65–98

Prompt 04 cold audit. French source edition and independent English translation.

Witness and editorial treatment

The controlling witness is the attached exact-scope PDF, leaves 3–36, corresponding to full-volume PDF pages 82–115. The original R27 and R28 bytes remain preserved as unverified prior work; only source-replayed derivative text is accepted. Pages 81–98 are transcribed directly from the authority. Pages 63–64 are copy matter, not author prose.

The French preserves the printed wording, notation, quoted statements, display grouping, and page seams. The English preserves all arguments and mathematical relations, including the legible source problems described below. No conjectural repair is inserted into either text. Running heads and the imprint are translated in the English reader. The author’s note remains a footnote; editorial comments appear only here.

P. 66: unqualified converse

In the paragraph beginning *Il est facile de voir*, the print states a converse from solvability for every prime factor of B to solvability for B itself. The unrestricted statement is retained. It needs qualification when repeated prime factors occur: for example, $A = 3$, $B = 9$ gives a solution modulo the prime factor 3, but no solution modulo 9, whose fourth-power residues are 0, 1, 4, 7. This counterexample is an editorial calculation, not text supplied by the witness.

Authority: exact-scope leaf 4 / full-PDF page 83, second paragraph.

P. 70: the printed plus sign

In the paragraph beginning *Le produit*, the print calls K a divisor of $\varphi'^2 + 2u'^2$. The plus sign is legible. Earlier on the same page, the factorization uses $\varphi'^2 - 2u'^2$. The plus is reproduced in both editions. The surrounding argument indicates a likely printed error; that diagnosis is editorial and does not change the diplomatic reading.

Authority: exact-scope leaf 8 / full-PDF page 87, the paragraph before the quoted theorem.

P. 71: restoration of the author's note

The inherited formula $r = \rho^2 \mp 1$ is an algebraic rewriting, not the printed expression. The witness reads $r = b\sigma^2 \pm 1$, followed by $s = \rho\sigma$. The two signs reduced to $+$ in R27's norm identity are also printed as $\pm$:

$$(rt \pm bsu)^2 - b(ru \pm st)^2 = p.$$

Both editions restore the printed forms. The source marker is $*$), attached after *impair* (*odd* in English), before the sentence's full stop. R27's extra closing parenthesis in the French body is not present in the source. The book title *Théorie des Nombres* is italic in the source; it is rendered as *Theory of Numbers* in English. The citation's printed numbering is retained.

Authority: exact-scope leaf 9 / full-PDF page 88, complete footnote.

Cold-audit detail witness: [qa/s04/source/p071_footnote_detail.png](#).

Other page records

Pages 65, 67–69, and 72–74 require no conjectural emendation. The separate pagewise apparatus files record the title and rules, the lexical reading *de théorèmes* on p.68, lower-case *d'où* on p.69, the equation labels and repeated factor arrays, and the exact continuation points. The cumulative catalogue has one record for each of the thirty-four author pages independently re-read in the cold audit.

The sentence at the foot of p.67 resumes at the head of p.68; that at the foot of p.71 resumes with the result labelled (α) on p.72. The reciprocity display announced at the end of p.72 begins p.73. The substitution sentence at the foot of p.73 resumes with its power of u on p.74. No bridging prose is supplied. In the final product on p.72 and the product near the head of p.73, the print stops after $hh'h''$ without another ellipsis; both editions retain that local punctuation rather than regularizing the lists.

Section 4 begins on p.74. Prompt 01 ended after its two-by-two factor display; Prompt 02 resumes with the proof that E and F are relatively prime on p.75. Superseded page sources and prior evidence remain preserved byte-for-byte. The reciprocity row at the foot of p.86 is followed on p.87 by its product. The integrated readers cover author pp.65–98 exactly once. The final cold collation replays every source leaf, including the inherited regions, rather than relying on previous acceptance.

Layout record

The readers are re-typeset, not photographic facsimiles. Original page boundaries, paragraph starts, display membership, formula punctuation, and the author-note anchor are retained. Physical within-page line wrapping and justification are not copied; line-end word division is reflowed, without changing lexical hyphens such as *tout-à-fait* and *à-peu-près*. These layout choices are explicit, not silent lexical normalization. The untouched authority and page images remain the reference for the original lineation and letterforms.

P. 77: the printed variable t

The paragraph beginning *En combinant ce résultat* says that the outcome depends on whether t has the form $8n + 7$ or $8n + 3$. The letter is visibly t , not b . Both editions retain it. The preceding equality is $\left(\frac{t}{b}\right) = \left(\frac{2}{b}\right)$, while Theorem I on p.78 gives the corresponding alternatives in terms of b . This discrepancy is recorded only here; no substitution is made in the author text.

Authority: exact-scope leaves 15–16 / full-PDF pages 94–95.

P. 82: the printed square on p

In the first paragraph, following *On voit, par l'équation*, the source reads $p^2 = \varphi^2 + \psi^2$. The superscript 2 is legible and is reproduced in both editions. On p.81 the decomposition is introduced as $p = \varphi^2 + \psi^2$. The extra square is likely a printing error, but that diagnosis is editorial, not a warrant for changing the diplomatic reading.

Authority: exact-scope leaf 20 / full-PDF page 99, first paragraph.

P. 85: ps , not ps^2

The parenthetical replacement equation reads $t^2 + au^2 = ps$. There is no superscript on this s , whereas the two immediately preceding equations both have ps^2 on the right. The printed difference is retained in both editions without supplying a conjectural exponent.

Authority: exact-scope leaf 23 / full-PDF page 102, the long methodological paragraph.

Pp. 75–86: page records and seams

No conjectural emendation is required on pp.75–76, 78–81, 83–84, or 86. The source equation labels (δ) , (δ') , (ε) , (ζ) , (η) , (θ) and (α') retain their local scope; the renewed (α') on p.86 belongs to the *Addition*. No renumbering is imposed.

The phrase ending p.77 continues with *de b* on p.78. The factor list at the foot of p.79 ends with h , and resumes with $h', h'', \dots$ on p.80. The two sign cases on p.80 are joined in (η) on p.81. The sentence at the foot of p.83 continues with *même chose* on p.84. The *Addition au mémoire précédent* heading and both adjacent rules are on p.85; its opening sentence continues with *simplifiées* on p.86. No bridging prose is inserted.

There are no new author footnotes on pp.75–86. The note mentioned on p.86 refers to the note near the beginning of section 3 on p.71, preserved in the cumulative readers. Source signatures 10*, 11, and 11* occur on pp.75, 81, and 83 respectively; p.81 also carries the volume imprint.

Pp. 87–98: the concluding pages

These twelve pages are transcribed directly from exact-scope leaves 25–36, full-PDF pages 104–115. No inherited transcription supplies their text. No conjectural emendation is introduced. The seven new local labels (β') , (γ') , (δ') , (ε') , (ζ') , (η') , (ϑ') and their references remain as printed; in particular, the renewed primed labels in the *Addition* are not renumbered to avoid earlier repetitions. The source’s open theta form is represented by ϑ in these new labels.

The source uses *impairment pair* on pp.87 and 93; the translation retains the term as *oddly even*. In these passages it means twice an odd number, as the source’s accompanying conditions $\nu = 1$ and $\mu = 1$ indicate. This explanation belongs to the apparatus, not the author text.

On p.90 the comparison sign has two equality strokes beneath the greater-than stroke and is represented by $\geq$. The factors b^{2k+2} , b^{2h+1} , b^{k-h} , and b^{h-k-1} , and the subsequent reuse of h in $g^2 + bh^2$, are retained. The two-by-two factor arrays on pp.88 and 94 preserve the printed order; no symmetric rearrangement has been substituted.

P. 89: the printed divisor b

In the opening paragraph, the print says that ψ is not divisible by b . It does not print δ at this point. Both editions now reproduce b ; the S03 transcription’s δ was an unmarked contextual repair. The surrounding argument concerns a supposed common divisor δ of E and F , and the next sentence again names δ . Thus δ appears to be intended, but this is an editorial inference, not the diplomatic reading.

Authority: exact-scope leaf 27 / full-PDF page 106, opening paragraph, third prose line. Detail: [qa/s04/source/p089_divisor_detail.png](#).

Continuities and source furniture

Seams 74/75 and 80/81 were rechecked against the scans: the first continues the factor proof, and the second combines its two sign cases in (η) . The 86/87 seam continues the prime-factor reciprocal-symbol row with its product, without a new heading or an invented connecting sentence.

The French source divides *der-nière* across 90/91: the page records retain *der-* at the end of p.90 and *nière* at the start of p.91. The English breaks at the corresponding phrase, *the last / equation*. Other incomplete sentences pass through the original page boundaries; the theorem announced on p.93 is the labelled statement at the head of p.94. Pages 97/98 continue the sentence about the second case. No new author footnote occurs on pp.87–98.

Signatures 12, 12*, and 13 occur on pp.89, 91, and 97. The volume imprint occurs on pp.89 and 97. The extended running head and the final horizontal rule on p.98 are retained in both readers, with the running head translated in English. All closing examples for $a = 5$ and $b = 3$, and both alternative pairs of equations, are included. No terminal word or further text is supplied after the rule.

Boundary and cold audit

Full-volume PDF page 115 is visibly printed p.98 and ends Paper IV. Page 116 is the next paper's title leaf and is inspected only to confirm exclusion. It supplies no text to these editions. The immutable full-volume file and the separately labelled boundary inspection image are provenance evidence, not additional reader content.

The catalogue now has thirty-four author-page entries; title leaf⁶³ and blank verso⁶⁴ remain only in the copy-matter and topology records. The retained source issues on pp.66, 70, 77, 82, and 85 remain literal. The additional *b* reading on p.89 is likewise preserved, with the distinction from contextual inference made above. The cold audit independently read every French line and formula and checked every English paragraph against the authority; its pagewise observations and concrete corrections are preserved separately.